Daily Drilling Report
SPD19B-03
8/31/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
35.0	12 1/4"	180.00	72.83	31.70	99.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
13 3/8" Shoe, 1,487.00mKB	220,877.90	28,562.21	8,164,824.17		30.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
2,600.00	2,694.00		94.00	19.15	4.9

Operations at Report Time
Drilling @ 2708m.

Operations Summary
Drilling 12 1/4" hole section f/ 2600m to 2694m.

Operations next 24hrs
Cont' drilling 12 1/4" hole to TD (+/- 2726m) . Sweep hole. Perform wiper trip to 2437m. Run back to bottom. Sweep hole. Spot weighted pill. POOH to surface.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Vahab Kari	Night Drilling Sup.
Hamed Daneshkhah / Moslem Yarahmadi	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/23/2014	10/13/2014
BOP Test	8/20/2014	9/11/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/23/2014	10/13/2014
H2S Drill	8/31/2014	10/1/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	6/16/2014	9/15/2014
Pit Drill	8/25/2014	10/10/2014
Trip Drill	8/28/2014	10/18/2014

STOP Cards

Number of Reports: 28

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:00	24....	12 1/4"	D	03	P	Directional drilling 12 1/4" hole section in rotation mode f/ 2600m to 2694m w/ 800 gpm/ 2150-2300 psi/ 37-40 kips/ 40 rpm/ 12-28 TRQ. ** Drilling with NL/NG. ** Pump 20 bbl Hi-Vis pill when STD down. B/R and ream down each stand drilled once. ** Take survey before connection. ** Gradually increase MW f/ 11.7 to 12 ppg before entering Aghar shale. >> Drilling procedure in Aghar shale member (Prognosis: 2677m - 2705.5m): * For the first 10m in Aghar, B/R gently to top of Aghar every 1m drilled. * For the rest of drilling in Aghar, B/R gently to top of Aghar every 5m drilled. * At stand down, pump 30 bbl Lo-Vis + 30 bbl Hi-Vis pill, B/R and R/D entire stand twice. * Spot 20 bbl 15 ppg weighted pill around BHA prior to connection. * Skip survey in Aghar. >>>> Drilling Mud Note : While drilling at 2610m. MI mud eng. reported significant increase in API F.L f/ 3.5 to 5 cc/30min. Decided to reduce the FL before reaching Aghar shale. Had to consume & add excessive amount of Starch (177 sxs) into the system. Due to poor quality of starch, the F.L remained constant at 5 cc/30min. Had to add Pac-LV (58 sxs) to help decrease F.Loss that lead to an increase in funnel viscosity f/ 62 to 78. Decided to add spersene (20 sxs) into the system to mitigate the effect of Pac-LV on viscosity. In order to minimize the effect of relatively high F.L (5 cc/30min) on the formation shale, decided to raise the KCL (7 BB) concentration f/ 5 to 7 %. Since the above mentioned materials are all consumed unplanned, MI cannot make any charges for these chemicals.
00:00	00:00						Sep 01 , 2014 00:00 - 06:00 Cont' directional drilling 12 1/4" hole section in rotation mode f/ 2694m to 2708m w/ 800 gpm/ 2150-2300 psi/ 35-40 kips/ 40 rpm/ 11-27 TRQ. ** Last Sample @ 2700m: Anhy (20%) + Dol. (30%) + Clst (50%)

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Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
2,694.00	KCl-Polymer Mud	12.00	66	28.0	29.000	3.8
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
9.8	0.200	1.800	0.800			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
12	14	2.0	76.0	22.0	55,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
13.000	23.000	29.000	2.6/97.4	68.0	2082.0	1505.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
5215.0	147.0	0.0			38.83393	0.28

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.10	5.	45	2.0	996.0	33.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Fadak-20	Crew Boat	8/31/2014 21:15	8/31/2014 21:40	Disemb 07 PAX and Emb 02 PAX for Kish. C/Off and departed for Sahar-2.
Marina-1	Supply Boat	8/16/2014 15:15		Stand by Boat.
Swissco	Supply Boat	8/31/2014 02:30		Delivered 05 Pallets of MI Chemicals from Sahar-2. Stand by anchorage.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Barite Bulk	28.0	249.0	162.0
Caustic Soda 2014	12.0	340.0	180.0
PTS (Temp. Stabilizer)	3.0	19.0	14.0
MI-Cide	4.0	15.0	5.0
KCL	7.0	52.0	21.0
Spersene	20.0	145.0	125.0
Soda Ash 25 Kg	64.0	550.0	227.0
CaCo3-C 25Kg	10.0	40.0	0.0
Starch	177.0	1,080.0	97.0
Pac-LV	58.0	422.0	97.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Barite BB	0.0	0.0	0.0
Cement G	0.0	0.0	179.0
Bentonite Bulk	0.0	0.0	8.0
Bentonite BB	0.0	0.0	0.0
Diesel Fuel	14.0	50.0	199.0
Barite Bulk	28.0	0.0	162.0
Drill Water	14.0	0.0	300.0
Potable Water	20.0	20.0	102.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
9	12 1/4	Varel	VM616PGUX-V2	6011971	PDC	M422	12/12/12/12/12/12/12/12/12
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.99	47.15	406,545.60	2,437.00	2,694.00	257.00	5.5	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
750	810	1,650.0	2,250.0	7	40	125	157
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
63.40	47.15	100.00	0.00	400	355	296	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
2,600.00	2,694.00	19.15	24,000.0	800	40	2,250.0	40	355

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWDP	5 7/8	4.000	27.83	231.39
7" Drilling Jar (COR-3022-AD)	7	2.938	5.85	203.56
16 jts 5 7/8" HWDP	5 7/8	4.000	149.23	197.71
X/O	7	2.813	0.79	48.48
1 jt 8 1/4" DC	8 1/4	2.810	9.46	47.69
8 1/2" Float Sub	8 1/2		0.79	38.23
8" NMDC	8	3.500	9.44	37.44
8" TM - Sub	8		2.03	28.00
8" NMDC	8	3.500	9.46	25.97
11 7/8" S. Stabilizer (COR3103AD)	11 7/8	2.813	2.16	16.51
8" PNMDC	8 1/16	3.500	4.03	14.35
9 5/8" Motor (AKO: 1.00° , Sleeve: 12 1/8")	9 5/8		9.95	10.32
12 1/4" PDC bit	12 1/4		0.37	0.37

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)
1	2,666.00	14.00	20	300.0	97	114
1	2,666.00	14.00	30	345.0	97	170
1	2,666.00	14.00	40	420.0	97	227
2	2,666.00	14.00	20	300.0	97	114

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No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)
2	2,666.00	14.00	30	340.0	97	170
2	2,666.00	14.00	40	408.0	97	227

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.35	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
2,616.26	248.00	21.00	2,478.56	-326.93	-711.26	782.80	0.00	0.00
2,645.20	248.20	20.90	2,505.59	-330.79	-720.86	793.13	0.13	-0.10

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
S5		2,571.60	2,436.87
S4		2,594.00	2,457.78
S3		2,622.00	2,483.92
S2		2,632.50	2,493.73
S1		2,675.70	

General Remarks

Comment

>> Requirements: (70 BB CaCo3 f/ MI MR #13) + (MI MR#14)

>>> Sent: NIL

>>> Received:

** Swissco f/ SHJ: (50 MT FO)

** Swissco f/ Sahar-2: MI Chemicals: (3 pallet Starch) + (2 pallet Pac-UL)